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REVIEW ARTICLE

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Demand-responsive transportation (DRT) in rural towns: the role of autonomy and perceived accessibility in inclusive and sustainable mobility

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ABSTRACT

This study investigates user perceptions of a demand-responsive transportation (DRT) service in a rural Swedish context, with a focus on autonomy and accessibility. It emphasizes the user experience alongside the service provider perspective to assess alignment between their goals and user experiences. Semi-structured interviews with users and providers revealed five main themes: service choice, user experience, perceived accessibility, perceived autonomy, and technology, with accompanying sub-themes. Users valued the flexibility of the X-line service, though they raised concerns about limited operating hours and challenges with technology, especially for older adults. Providers highlighted the benefits of on-demand booking and real-time updates but acknowledged issues such as longer wait times and capacity limitations that affected satisfaction. The study highlights gaps between user needs and provider intentions, particularly related to age and technological proficiency. While DRT services offer improved autonomy and accessibility over traditional public transport, limitations in flexibility, wait times, and reliance on digital platforms remain. The study concludes that the success of DRT services depends on user readiness and operational efficiency, with implications for the design of inclusive and sustainable mobility solutions in rural and small-town settings, and calls for further research to explore variations across geographic and cultural contexts.

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1. Introduction

Transportation and the way people travel on a daily basis is changing due to technological advancements, changing consumer preferences, and global

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sustainability goals (Deka et al., 2023; Saif et al., 2024). In this context, flexible public transport and shared mobility are increasingly positioned as tools for achieving sustainable mobility, reducing car dependency, lowering emissions, and improving social inclusion, especially in places where fixed-route systems are inefficient. Traditional fixed-route transportation systems are increasingly challenged by more flexible and personalized travel solutions which can be facilitated through innovative platforms and digital technologies. This transformation puts new emphasis on user autonomy in travel, the ability to make independent choices about when, where, and how to travel. Unlike rigid transit schedules, these emerging travel solutions give users greater control over their travel decisions (Bina & Biassoni, 2023). Shared mobility services such as ride-hailing, bike-sharing, and micro-mobility solutions are redefining mobility (Nikitas, 2024) by offering flexible alternatives and reducing the need for private car ownership. Beyond their operational flexibility, these services also reshape the experience of mobility by enhancing autonomy, allowing travelers to make more independent travel decisions about when and how they move, thus reinforcing a sense of self-governance in everyday travel (Bina & Biassoni, 2023). In parallel, the growing reliance on digital platforms and algorithm-driven systems has further amplified this trend, enabling transportation services that respond dynamically to user needs rather than adhering to fixed schedules. This shift is facilitated by new transportation services, which, leveraged by digital platforms, offer on-demand, convenient, and cost-effective alternatives to conventional modes of transportation.

While shared mobility has significantly influenced urban transportation systems, rural areas present distinct challenges. Their lower population density and longer travel distances require specific adaptations—adaptations that are increasingly acknowledged but remain comparatively underexplored in the literature. While Zhu et al. (2023) highlights the role of technology in enhancing urban mobility and sustainability, rural systems need to be considered within a different operational context, where issues such as infrastructure limitations and service coverage over larger geographic areas take center stage. Because rural residents often face higher car dependence, limited service frequency, and longer access distances, the sustainability and livability benefits of flexible and demand-responsive public transport services in rural contexts depend not only on system performance but also on whether services are experienced as usable, reliable, and enabling of everyday participation.

A relatively new urban transportation service that challenges traditional fixed-route transportation systems is demand-responsive transportation (DRT) (Abdullah et al., 2021). Unlike conventional transportation systems, DRT adopts a responsive approach based on user demand, offering not only increased accessibility but also greater perceived autonomy in travel. DRT was initially targeted at specific demographic groups, such as individuals with impairments, but

recent developments have seen broader adoption of these services for the general public. DRT can potentially emerge as a viable transportation option in urban and rural areas (Berrada & Poulhès, 2021; Schlüter et al., 2021; Velaga et al., 2012). This is consistent with previous research showing that DRT services tend to perform best in sparsely populated areas where fixed-route services are insufficient or unsustainable (Brake et al., 2007; Dytckov et al., 2022; Heinitz, 2022; Schofer et al., 2003).

While previous research has predominantly concentrated on studying the "hard factors" associated with DRT, such as system-level metrics, operational efficiencies, policy, and regulation (e.g. Dawda et al., 2021; Klopp, 2021; Pettersson, 2019; Rahimi et al., 2018), a growing body of work has also begun to explore user experiences and perceptions. Building on this research, the present study places particular emphasis on users' perspectives of travel autonomy and perceived accessibility, aspects that are pivotal for understanding how DRT services are experienced in everyday mobility. However, although accessibility has been widely studied in transport research, perceived accessibility in DRT contexts and how it relates to service design features such as booking, reliability, and operating windows is less clearly synthesized. There is a growing interest among practitioners in user perspectives of DRT in rural and semi-urban contexts, which is also reflected in recent publications studying DRT services around the world (Schasché et al., 2022). In Austria, research has shown that satisfaction with both trip-level experience and the system design of the DRT affects overall travel satisfaction, and proposes that attention also should be given to different user segments (Gkavra et al., 2024), while another Austrian study suggests that urban and rural DRT are surprisingly similar from a user perspective (Schilder et al., 2024). In a case in England, researchers observed that trips were primarily non-work related, that planning should be taken for extra capacity in peak hours, and importantly, that most users that made one trip continued using the service (Zahedi et al., 2024), and in Germany, older people were observed as more satisfied, open-minded and more positive than expected (Kersting et al., 2021). In their review of DRT studies, service access quality, trip costs, service access, information, and the booking service were identified as main influencers of travel mode decisions, while socio demographics such as age, gender, income, and access to car provided inconclusive results (Schasché et al., 2022).

Previous studies have also shown that drivers play an important role in user satisfaction. Nelson and Phonphitakchai (2012) found that drivers' friendliness, helpfulness, and local route knowledge were key contributors to positive service perceptions. This also aligns with earlier observations by Cass et al. (2003), who described drivers as crucial intermediaries between operators and passengers. These interpersonal factors can foster trust and continuity in usage, suggesting that DRT experiences are co-created not only through digital booking systems or service design, but also through human interaction at the point of delivery.

While knowledge from the user perspective is gaining attention, the present study extends the existing body of research by also integrating perspectives from managers responsible for DRT service provision. This dual perspective—combining user and provider viewpoints—offers a more nuanced and comprehensive understanding of how DRT systems are perceived and experienced. By examining how managerial efforts to enhance autonomy and accessibility align with user experiences, this research not only addresses a critical gap in the literature but also contributes practical insights for improving DRT design and implementation. In addition, we explicitly position autonomy in terms of users' perceived control over travel choices, timing, and means of travel (what/when/how), together with perceived accessibility, as experiential determinants of how DRT is evaluated and used, thereby linking everyday user experience to discussions on sustainable mobility. In doing so, this study not only highlights the importance of aligning operational goals with user needs but also advances the discourse by demonstrating how these factors can collectively contribute to sustainable and inclusive mobility solutions.

To explore these aspects, we report on a case study involving the introduction of a DRT system within a small Swedish rural town called Säffle comprising around 15,000 inhabitants. In December 2022, the Säffle municipality launched the DRT system "X-line" (X-linjen, in Swedish) as a temporary test service replacing the traditional bus services serving the rural area around the municipality. The "X-line" quickly evolved from its initial pilot phase, securing a permanent place in the town's transportation ecosystem. With its aim and case focus, the remainder of this paper is structured as follows: First, we discuss relevant research focusing on travel autonomy and perceived accessibility from a user perspective. This section is followed by an overview of the research design, including detailed information on the X-line case. The presentation of the findings is followed by the final section which contains the discussion of the findings and recommendations for future work.

2. Travel autonomy

Autonomy, defined as the individual's capacity for self-governance, involves the freedom of choice and authorship of actions (Hmel & Pincus, 2002; Ryan & Deci, 2006). Acting according to one's volitions has been linked to need fulfillment and goal-pursuit, suggesting that autonomy also has an important behavioral component (Deci & Ryan, 1987). Ryan and Deci (2000) linked autonomous actions to intrinsic motivation, suggesting that individuals have an innate desire to fulfill goals that are in line with their core values and desires. Behaviors that are autonomous, and supported by the environment, are thought to promote personal achievements and psychological well-being. On the contrary, behaviors that are controlled by external factors or persons can have the opposite effect by restricting individual choice and control. Such thwarting of inner

psychological needs can in the long run have a detrimental impact on intrinsic motivation and overall well-being (Ryan & Deci, 2000).

To explain how autonomy functions in everyday decision-making, Zhang et al. (2023) developed the functional model of autonomy to explain how autonomy is linked to three main decisions that individuals are continuously challenged by. The *what* decision refers to a goal that a person is going to achieve in a given situation. The *when* determines the timing to perform the target behavior (i.e. when to act to achieve the goal), and the *how* refers to the means through which the behavioral goal is achieved. In the context of transportation, autonomy is shaped by these three factors. For example, a person deciding to travel to attend a social event (the what), choosing a bus (the how) and setting the time to make it on time (the when) illustrates how these decisions shape *travel* autonomy. However, external factors like an unreliable bus schedule can limit control over these decisions, potentially affecting future choices and diminishing travel autonomy. This model highlights how disruptions in decision-making can reduce motivation and travel autonomy.

Few studies exist that have been carried out on autonomy in the context of transportation. Márquez et al. (2019) find that individuals with mobility impairments face challenges in both autonomy and perceived accessibility due to transportation-related limitations, particularly in wheelchair accessibility. Thus, it seems that personal autonomy and perceived accessibility are two related concepts that affect quality of life as they can hinder choice. This relationship has been shown by studies investigating transportation services for people with physical disabilities and finding that limited access to transportation affects well-being, such as social participation (Bascom & Christensen, 2017). Another study found that mental illness, as another form of disability, relates to travel autonomy and perceived accessibility negatively (Friman & Olsson, 2023). Both autonomy and social participation are integral aspects of intrinsic motivation and psychological well-being, in which transportation seems to play an important role (Ryan & Deci, 2000). Using the functional model of autonomy, it appears that the "what" to do (the choice) is hindered by the "how" to get there, which in turn affects the way an individual perceives autonomy.

In transportation research, autonomy has been studied under the term "decisional autonomy" (Meurer et al., 2014). For example, with a focus on older adults' mobility, Meurer et al. (2014) find that decisional autonomy for older people is the ability to plan for oneself *how* and *when* to undertake journeys of whatever kind. This qualitative study linked decisional autonomy with car possession since driving means having the option of being mobile whenever one wants and needs it, thus allowing immediate wishes to be put into action without the need for any detailed travel planning. On the contrary, ridesharing is perceived as autonomy reducing, as rides are not predictable with respect to time and route and therefore give rise to uncertainty (Meurer et al., 2014). This link supports Zhang et al. (2023) ideas on how restricting the agent (i.e. the decision-maker)

in any sense can infringe on the individual's sense of autonomy and harm psychological well-being. Hence, the *"what"* decision (satisfying a particular need; e.g. traveling into town) is impeded upon by the *"when"* (not being able to perform the target before due to dependency on others or external factors).

Decisional autonomy is of particular interest for the current study. If car ownership represents complete autonomy and public transport is the most restrictive form of autonomy, we are interested to see if and how an option that closely mimics the advantages of car ownership can support autonomy. While demand-responsive transportation shares similarities with ridesharing, such as dependency and a degree of uncertainty, there are reasons to expect different findings on autonomy. Firstly, on-demand transportation is designed to not infringe upon the "*when*" decision since individuals choose for themselves when they want to travel and can take control by selecting when to travel through a phone application. This is because on-demand transportation services are designed to be flexible to choose rather than restricted to a set schedule. This should, in theory, support autonomy. Secondly, most research on autonomy in relation to transportation has been carried out among the elderly and disabled. These population groups may interpret mobility and autonomy differently than those whose independence is not restricted by physical or mental impairments. This is because being aware of one's limits and the fact that autonomy may diminish in the near future may enhance the need to preserve autonomy for as long as possible. The above arguments are, however, only speculative and have yet to be studied in more depth. In line with this research need, the present study is designed to explore the experienced autonomy by users of an on-demand service. By additionally adding the service provider perspective, the study further investigates if the view on autonomy aligns between service providers and users.

3. Perceived accessibility

In contrast to autonomy, transport research has often focused on accessibility. This focus is not surprising, given that the possibility to participate in activities outside the home is essential for living a meaningful and satisfying life, whereby transportation plays an important role. When a transportation system does not align with one's mobility needs, it can negatively affect quality of life.

Traditionally, emphasis has been placed on objective accessibility measures, such as proximity and travel time to predefined destinations (e.g. schools, healthcare, workplaces), and many transportation planning initiatives rely on such metrics. Yet there are concerns about whether these objective measures provide a complete picture, as individuals do not necessarily view accessibility in line with these calculations (Curl et al., 2011; Lättman et al., 2016; Pot et al., 2021). Several studies have shown discrepancies between objective measures and perceived accessibility (e.g. Lättman et al., 2018; Pot et al., 2023). For

instance, a bus stop may be objectively close and accessible, but the individual may not perceive it as accessible due to feeling unsafe. The opposite may also be true: rural areas that objectively have fewer opportunities may not be perceived as non-accessible by those living in the area, perhaps due to the ease of accessing nearby opportunities by car, or that people in dense areas are less sensitive to travel time, number of opportunities, and distance (Pot et al., 2023).

As objective and perceived accessibility have been shown to differ, scholars have studied other factors of relevance for perceptions of accessibility, including experiential factors such as convenience, safety, reliability, time, and cost (e.g. Lättman et al., 2016; Olsson et al., 2021; Sukhov et al., 2023; Sukwadi et al., 2022; Wang et al., 2022; Yarlagadda & Gupta, 2025). For example, Curl et al. (2011) and Lättman et al. (2016) demonstrated that individual perceptions of accessibility are more strongly influenced by subjective experiences such as safety, comfort, and reliability than by purely spatial proximity. Studies in Sweden and Russia (Lättman et al., 2018; Lukina et al., 2021) show that proximity to the city center is not important for perceived accessibility. This contrasts with objective assessments of the same areas where distance and travel time to the city center constitute a significant part of the evaluation. Likewise, a study conducted in the Netherlands finds no significant differences in perceived accessibility between urban and rural residents (Pot et al., 2023). Specifically, people living in rural areas reported that proximity to activities was not as crucial as for those residing in urban areas. A compensating factor for the lack of local activity proximity in rural areas appears to be the use of private cars, signaling a strong car dependency in these settings. With a car, one is able to reach all activities, contributing to a high perceived accessibility and autonomy (Meurer et al., 2014). It has also been shown that those with high car usage believe that restrictions on freedom to use the car (e.g. bans on using the car on certain days or in specific areas) would significantly reduce accessibility, regardless of where they live (Lättman et al., 2020). Maintaining accessibility while restricting car usage to promote a more sustainable transportation system will hence likely be challenging, especially in less densely populated areas.

While autonomy is understood as being able to exert control, perceived accessibility can be viewed as a way to achieve one's goals, which may also in part be reflected in the functional model of autonomy (Zhang et al., 2023). The agent will not be able to make the *how* decision without a perception of the goal as accessible. As shown in previous research, autonomy and perceived accessibility seem inherently linked (Friman & Olsson, 2023; Ryan & Deci, 2000). Or put in specific terms, the more autonomy experienced, the more accessible an opportunity will be perceived—as it is in the hands of the individual decision-maker, rather than an external entity. Building on this body of work, the present study aims to explore how perceived accessibility and travel autonomy are experienced in the context of a demand-responsive transportation (DRT) service trial in a rural town. Furthermore, we extend previous studies by

adopting a dual-perspective approach, investigating both the users' and service providers' perceptions of a demand-responsive transportation (DRT) service. With particular attention to how service design features, such as responsiveness and booking flexibility, interact with users' sense of autonomy and accessibility, this approach allows us to contribute to a more nuanced understanding of accessibility that goes beyond spatial metrics and incorporates the lived experiences and motivations of rural travelers.

4. Research design

In this section, we provide an overview of the selected case study and we outline the specifics of our data collection methods and analysis procedures.

4.1. Case study

This study used the small Swedish town Säffle as a case study. This case is intriguing because it describes a trial of a demand-responsive transportation (DRT) system in a rural town. Specifically, on 11th December 2021, local transport authorities in the town of Säffle implemented the DRT system the X-line as a temporary transport service in replacement of a regular public bus line, which was running on low efficiency and high costs. The X-line started as a pilot but then was established as a permanent service due to its success, including stronger demand and increased usage by local commuters. The DRT's travel development over the operating period is depicted in Figure 1. As seen, the number of travels varies over the years of operation, but the overall trend is positive.

To date, the X-line DRT system consists of two shuttle buses, each accommodating 15 passengers alongside with space for one wheelchair and a stroller. While the X-line covers the same geographical area as the previous bus service, it offers a more flexible solution with 152 virtual stops for passengers to choose from throughout the larger area around the town Säffle. X-line operates from Monday through Saturday without a fixed timetable, serving a broad demographic that includes children aged 0–6 years (18%), youths aged 7–19 years (14%), adults aged 20–64 years (63%), and seniors over 65 years of age (5%), demonstrating its importance across all age groups. Trips primarily revolve around commuting to school, work, shopping, and other errands, as well as using the service as a shuttle to other modes of transportation. Approximately 92% of all trips during the operating period were booked via a dedicated app, with the remaining 8% arranged by telephone.

The pilot phase of the X-line was evaluated as successful due to increased demand and usage of the service throughout the pilot (see Figure 1). Its success has also spurred the initiation of on-demand pilots in other regions of Sweden. For example, two cities in the north of Sweden have recently initiated similar on-demand travel systems after being inspired by the success of X-line

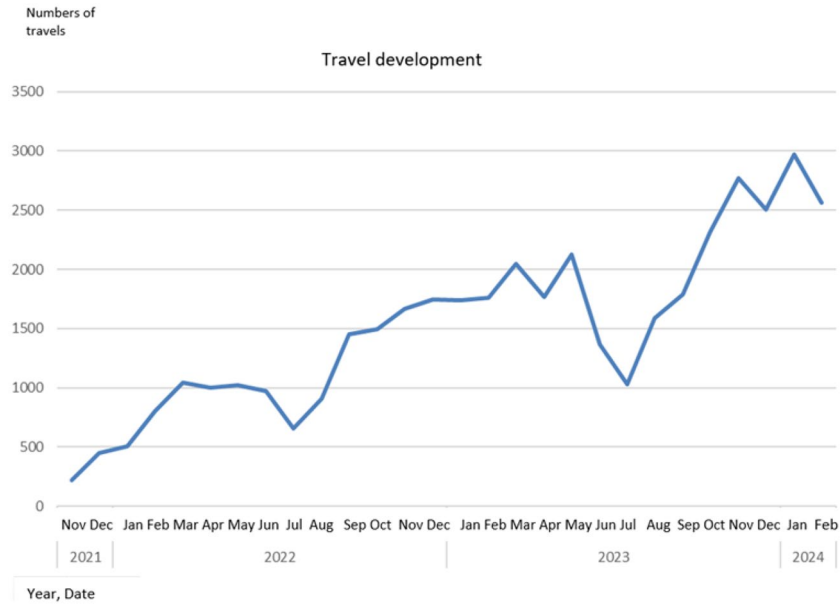


Figure 1. X-line travel development, specified by the number of travels over the operating period.

in Säffle, and two rural towns in the outskirts of Gothenburg are currently running similar pilots with their "Bus-on-Demand". After more than a year of operation in Säffle, the local transportation authority aimed to include travelers' experiences in their evaluation of the service, which enabled the present research.

4.2. Interviews

The study aimed to explore how users perceived a DRT replacing a conventional public transportation service with a focus on perceived autonomy and accessibility. In addition, we sought to capture the service provider's perspective on autonomy and accessibility to assess whether their intentions align with the user experience. This dual-perspective approach was chosen to provide a comprehensive understanding of how autonomy and accessibility are perceived by those who directly interact with the service and those responsible for its provision. Instead of seeking agreement among participants, the study focused on identifying both common themes and meaningful differences in their experiences, which reflects the core aims of qualitative research. In Figure 2, the research design and each step of the dual process are visually displayed.

To achieve our aim, we conducted semi-structured telephone interviews using a predefined interview guide. The guide included predetermined questions on autonomy and perceived accessibility while allowing flexibility for

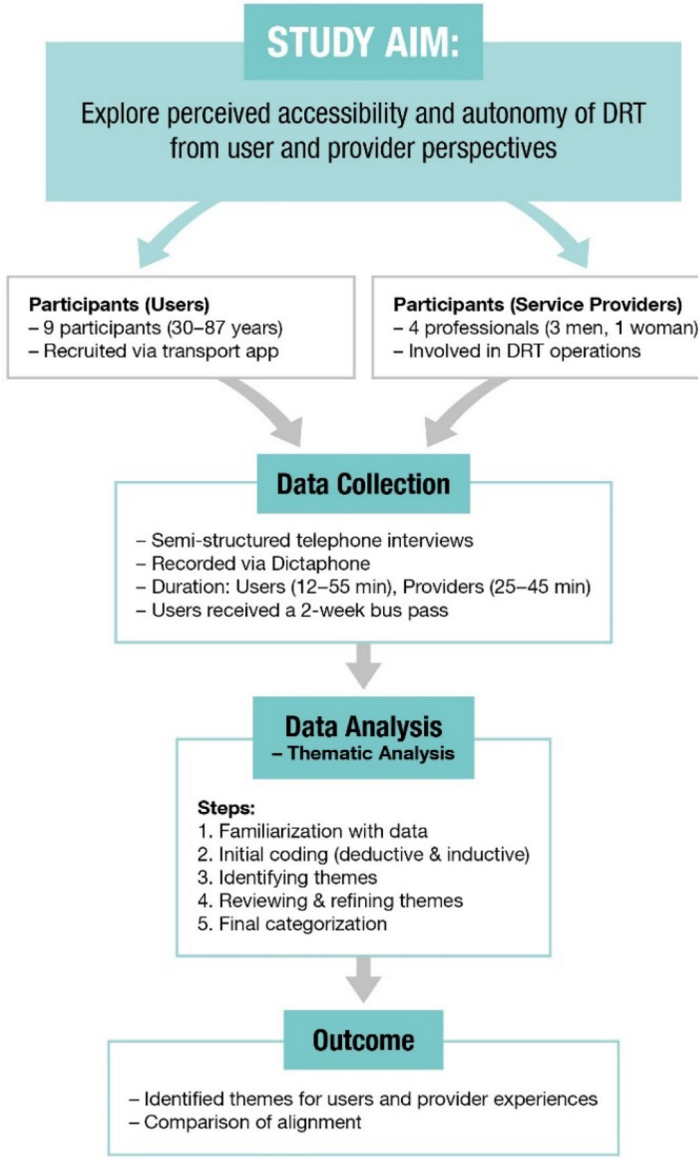


Figure 2. Research design.

open-ended discussions. The interviews were carried out by two members of the research team. Participants were adult users who had voluntarily signed up to take part in the study. This recruitment strategy was chosen to ensure that all participants had direct, first-hand experience of using the X-line demand-responsive transport service, which was essential for addressing the study's exploratory research questions. The interview date and time were mutually agreed

upon by the interviewer and interviewee, and participants were telephoned at the agreed time. The interviews started with an introduction re-familiarizing the participants with the study's objectives and ensuring their informed consent. Subsequently, basic demographic information such as age, gender, and living situation was gathered followed by more in-depth questions regarding their experience of using the service, with particular emphasis on perceived accessibility and autonomy. The interview guide was used consistently across all interviews; however, some respondents were more inclined to elaborate on their answers, leading to additional follow-up questions in some cases. This flexibility allowed participants to emphasize issues they themselves perceived as most relevant, thereby reducing the risk of overly constrained or leading interview responses.

All interviews were recorded using a Dictaphone and took between 12 and 55 min. Participants received a 2-week free bus pass as compensation for taking part in the research. The same procedure was carried out for interviews with service providers associated with X-line, but without the incentive. These interviews took between 25 and 45 min. Both samples are described in the following sections. All recorded interviews were transcribed by a third-party consultancy service. Both data sets were analyzed by following a thematic analysis procedure (Braun & Clarke, 2006). Data collection and preliminary analysis were conducted iteratively, allowing the researchers to monitor the emergence of themes across interviews.

4.3. User sample

Nine users agreed to participate in the study (5 women, 4 men; aged 30–87 years; see Table 1). A purposive sampling strategy was employed, meaning that all participants were current or former users of the X-line demand-responsive transport service. Recruitment took place via the local transport booking application, where an advertisement for the study was published, and interested users contacted the researchers directly.

Table 1. User demographics.

Participant	Gender	Age	Type of accommodation	Town of residence	Ridership frequency
A	Female	30	Apartment	Outskirts of town	Moderate
B	Female	30	Apartment	Säffle	Moderate
C	Male	75	Apartment	Säffle	Low
D	Female	41	House	Other city (summer residence)	High
E	Male	53	Apartment	Outskirts of town	High
F	Female	54	Apartment	Other city	Low
G	Male	73	Apartment	Other city	Low
H	Female	87	Apartment	Säffle	High
I	Male	31	Apartment	Säffle	Moderate

While the sample size is relatively small, it is appropriate given the qualitative and exploratory nature of the study. The aim was to generate in-depth, experience-based insights from individuals with direct engagement with the service, rather than to achieve statistical representativeness. Sample adequacy is determined by the informational richness and relevance of the data in relation to the research aim, and small purposively selected samples are well established when investigating lived experiences in depth (Bouncken et al., 2026; Hennink & Kaiser, 2022).

Data saturation was used as the primary criterion for determining sample sufficiency. Following Braun and Clarke's (2006) conceptualization, saturation was defined as the point at which no new themes, codes, or analytically relevant dimensions emerged from successive interviews. Data collection and analysis proceeded iteratively, with interviews coded sequentially. During the final interviews, all data segments could be accommodated within the existing coding framework, and no novel themes related to autonomy, perceived accessibility, or technology use were identified. This constituted a pragmatic stopping criterion for further recruitment.

The age range of participants was broad, reflecting variation in life situations and levels of transport dependence. Only two participants were aged over 60, which limits the extent to which the findings can be interpreted as representative of older users as a distinct group. Rather than providing a comprehensive account of older adults' experiences, the study offers indicative insights into how age and technology readiness may intersect with perceptions of DRT services.

Finally, the user sample does not fully reflect the demographic distribution of all X-line ridership. Our focus on adults omitted younger age groups, and recruitment via an app-based system may have excluded individuals with limited digital access or lower technology proficiency. Consequently, the findings should be understood as analytically transferable to similar rural demand-responsive transport contexts rather than statistically representative of the full user population.

4.4. Service provider sample

Four interviews were carried out with professionals involved in the operation of X-line (3 men and 1 woman). Three of these interviews were with representatives from the authority overseeing on-demand transport and overall mobility, while one was with a representative from the service operator. Due to the specialized nature of the study and the small pool of professionals with direct knowledge of the service, we believe that these four participants provided rich, essential insights into the service provider's perspective on autonomy and accessibility. Given the limited number of interviewees, demographic details are minimally disclosed to ensure their anonymity.

4.5. Qualitative research approach and analyses

Thematic analysis, which is a qualitative research method used for organizing and describing data by uncovering underlying themes, was used to analyze the interview data. This method involves the following key stages: becoming familiar with the data, generating initial codes, identifying themes, reviewing themes, and ultimately generating the final report (Braun & Clarke, 2006). To ensure coherence with the purposive sampling strategy and the iterative approach described in Section 4.3, data collection and analysis were conducted iteratively. Interviews were transcribed and coded sequentially, allowing the developing analysis to inform ongoing assessment of data sufficiency and theme development. This iterative process also enabled us to determine when thematic saturation had been reached, that is, when no new themes or analytically relevant insights emerged from additional interviews.

In the first stage of our study, we transcribed and coded the interviews conducted with users. This involved immersing ourselves in the data to gain a comprehensive understanding of its content. Subsequently, initial codes were generated from the transcribed interviews to capture meaningful segments of the interviews. This involves a process of systematically assigning descriptive labels or codes to segments of the data that capture key concepts, ideas, or recurring patterns. Codes were generated both deductively, based on our research aim and related theory basis, and inductively, allowing new themes to emerge from the data itself. The codes can be significant words or phrases. Once the initial coding was completed, codes were collated and grouped into potential themes.

Themes were identified based on shared patterns, meanings, or concepts across the data set. This process involved comparing and contrasting coded segments to uncover overarching patterns and connections within the data. The identified themes underwent a process of refinement to ensure clarity, coherence, and relevance. This involved reviewing and revising the initial themes, merging or splitting them as necessary, and ensuring that each theme accurately captured a distinct aspect of the data.

Once a coherent thematic structure was established for the user interviews, each theme was analyzed in depth by revisiting the associated data examples, examining contextual nuances and exploring variations within and across participants. This facilitated a comprehensive understanding of the meanings and implications embedded in the material.

The same analytic procedure was applied to the service provider interviews. To enable comparison between users and providers and to ensure analytical consistency across datasets, the coding framework developed from the user interviews was used as an analytical reference point. Provider interviews were coded inductively, but examined in relation to the user-derived themes, allowing areas of alignment and divergence to be identified. This approach maintained coherence across datasets while preserving the distinctiveness of each perspective. The final set of codes and themes is presented in Tables 2 and 3.

Table 2. Thematic analysis coding of interviews with users.

Main themes	2nd order coding	1st order coding	Illustrative quotes
DRT Service Choice	A necessity	Absence of car/ license	*I'm using the X-line because I don't have a car or driver's license.*
		Forced to give up the license	*I am visually impaired so I had to give up my license.*
	Choice and preferences	In the absence of other alternatives	*I would rather take the car, anytime. I really miss it.*
		Walking/exercise	*I used the X-line out of curiosity, but I didn't need it to get around.*
			My legs are healthy, so my primary mode of transportation in Säffle is by foot and it's an excellent way to get around.
DRT user experience	Internal	Expectations	*I had no expectations because I had never heard of something like this before… It was something I had never experienced.*
		Emotions and attitudes	*I felt it worked well and it was pleasant, so I didn't experience anything negative at all.*
	External	Bus drivers	*The service offered by the drivers is exceptional, like nothing I've experienced before. Very helpful and nice.*
		Reliability of service	*They should write that in the app, that the pick-up time is approximate, so that people know.*
Perceived accessibility	Objective	Good	*They stop pretty close every time, I rarely have to walk at all.*
		Poor	*They should extend their hours. On Sunday they don't operate at all, and on Saturday I think it's only for 3 hours.*
	Subjective/ perceived	Supportive/not supportive of needs	*It's given me the opportunity to take on more shifts at work.*
			The X-line does not support my transportation needs.
		Helps everyday life	*Before the X-line I felt very limited with what I could do, due to my social anxiety. Now with the X-line I get out much more.*
			It has helped me a lot this summer to get around since I don't have a car and everything is not within walking distance.
			The X-line hasn't changed my life but it has simplified it.
			I would reach my destination without the X-line, it would just complicate life a bit more, not having the service.
Perceived autonomy	Freedom and being able to choose when to travel	Flexible travel schedule	*Freedom within the hours that are being offered.*
			Not full freedom. There are some days when you aren't able to travel late, and one time it happened that they stopped operating without any notice, so you have to adapt.
	Importance of autonomy	Inconsistent service reliability	*When I have to go and ask my neighbor for a lift to take my bags to the recycling, that impinges on my sense of freedom.*
			I think it's pretty important if you suddenly have to travel somewhere, then the X-line caters for that.
	Factors affecting perception of autonomy	Impinges on autonomy	*Having to ask a neighbor for help with transportation*
		Strengthens need for autonomy	*If I need to be somewhere*

(*Continued*)

Table 2. Continued.

Main themes	2nd order coding	1st order coding	Illustrative quotes
Technology	Difficulties	Technological barriers for older users	*Older people have difficulties using the app.* *It was difficult to use the phone, so then I had to call in, but the people answering the phone has no knowledge of the X-line.* *It would be better and easier to be able to call the bus locally. I've heard many people say; how do I access that bus?*
	Opportunities	Ease of app use Desire for payment integration	*The app is super easy to use. It didn't take long to learn how to use it.* *It's easy to use but it is faulty. They have to fix many things with the app.* *The app is easy to use. My only wish is to be able to pay through the app as well…*

Table 3. Thematic analysis coding of interviews with providers.

Main themes	2nd order coding	1st order coding	Illustrative quotes
Provided autonomy	Flexibility and freedom	Whenever wherever Control of the journey	*The experience is that it is you who choose when the bus runs, and where it runs. This increased autonomy in travel decisions contributes to a more seamless and efficient travel experience for users.* *… it is the customers themselves who decide when the bus should run and where it should go.*
	Closeness	Outside the doorstep Goal-oriented support	*You have received good service and were picked up near where you were and dropped off close to where you were going.* *…this service can at least take you very close from where you are, to very close to where you want to go*
Drivers enhancing the DRT experience	Familiarity	Knowing the driver Same car same driver	*I think the key to feeling confident about sending your seven-year-old to football practice is knowing who the drivers are, you know.* *They also see who made the booking. It shows up in the booking, so they can say 'Hi Britta' or 'Hi Magnus.*
	Interaction	Mutual recognition Personal service	*Then you, well, you become friends with them. So it becomes a sense of security for parents who would let their children ride.* *Very engaged drivers. They think it's fun.*
Technology integration into the DRT service provision	Usability	Overall solution Easy to use	*But the latest that we have now during this month is that just over 85% actually book via the app; otherwise, you can contact Värmland public transport and their customer service to make the booking*
	Limitations	Work under progress Overall solution	*Then there are some opinions about not being able to pay for the trip directly in the app, instead, you have to do it separately, either in our other app or with a bank card. So there are some comments about that, but overall, I think people are generally satisfied with the app.* *The whole service needs further development. We see that there's potential to improve many parts of it.*

(*Continued*)

Table 3. Continued.

Main themes	2nd order coding	1st order coding	Illustrative quotes
The dynamic nature of the system	New travel patterns	Shorter and more trips	*We now serve other areas in Säffle that we didn't cover with the previous city bus.*
		Variety of travelers	*…so we've made life easier for parents in Säffle who might work on weekends or have more children at home.*
	Availability	Almost like a taxi service	*This type of transportation is much more useful for all kinds of errands compared to the route we had before.*
		Closer to you	*This is a step closer to matching the car.*

5. Findings

The findings are reported in three steps: first, in Section 5.1, we focus on the service users, secondly, in Section 5.2, we focus on the service providers, which is followed by Section 5.3 where we compare perspectives from service users and providers.

5.1. The user's perspective on the DRT service

The interviews with DRT service users led to five main themes, eleven 2nd order codes, and nineteen 1st order codes. The following five main themes were identified as part of the data analysis: (1) DRT Service Choice, (2) DRT user experience, (3) Perceived accessibility, (4) Perceived autonomy, and (5) Technology, with accompanying sub-themes. The codes and aggregated dimensions alongside examples of participant quotes are presented in Table 2. Below, we present the findings associated with the way service users perceived the DRT service. Three of the identified themes stems more or less from the interview guide (i.e. perceived accessibility, perceived autonomy, and DRT user experience, which is in line with the research aim), while technology and DRT service choice are themes derived from spontaneous comments related to user experiences.

The table above presents an overview of central quotations. The following sections offer a more detailed exploration of each main theme.

5.1.1. DRT service choice

Participants' reasons for choosing the X-line service differed across the sample, although all had chosen the X-line as a transportation mode due to the non-accessibility to a car. Some claimed the X-line was their only transportation mode and that they had a lack of choice, *"I'm using the X-line because I don't have a car or driver's license."*

For others, the X-line was one of several choices, with alternatives such as renting a car or walking/bicycling: *"I used the X-line out of curiosity, but I didn't need it to get around."* This suggests that the motive for using the X-line differed

on a need level; for some, it was a necessity, for others, it was a preference at a particular given moment but also among other alternatives. Similarly, another participant admitted that he would reach his destination without the X-line, but that it would take a bit more planning. This division on a need level poses an interesting dilemma – how can individuals in the "choice group" be motivated to select on-demand solutions in the long run even in the presence of other alternatives, such as the car.

5.1.2. DRT user experience

The experiences of the X-line were important for their overall perception and continuous use of the service. Since the X-line was a newly introduced transportation service to the local community, many tried the service with no prior expectations around the concept "on-demand". The ones interviewed reported positive and pleasant feelings, especially highlighting the absence of negative emotional experiences: *"I felt it worked well and it was pleasant, so I didn't experience anything negative at all."* Two factors were seen as central to shaping the DRT experience: (1) the interaction with the bus driver and (2) the reliability of the service.

Bus drivers were seen as an important contact person who contributes to passengers' overall perception of the DRT service. The majority of participants spoke of the bus drivers as an important part of their experience, with particular emphasis on their excellent service, flexibility, and friendly manner: *"The service offered by the drivers is exceptional, like nothing I've experienced before. Very helpful and nice."* One participant even mentioned that she was greeted by name as she entered the bus, which added value to her experience. The attitude and interaction with the driver seem to be highly valued. The smaller and more intimate size of the vehicles may explain why the driver plays a more significant role than in traditional public transportation services—and also to why the driver show commitment. This was strengthened in the discussions about the drivers' service and commitment. As resonated by one participant: *"But the two of them [the drivers] act as if they owned the company… they are very involved and show incredible commitment."*

The second key factor that contributed to the passenger experience was the perceived reliability of the service. Nearly all participants mentioned that a reliable service is essential, including short waiting times—especially when pick-up times are not met with 100% accuracy. As one participant mentions: *"If you book for 18:20, you can't be 100% sure that it'll arrive at 18:20… but still, I don't think it really matters that bloody much."* Here, the app plays a key role in specifying and updating pick-up times in real-time to avoid frustration. As one participant suggested, the app could include a note stating, "*The time is approximate*."

5.1.3. Perceived accessibility

The X-line users reported on aspects that relate to both objective (such as proximity to pick-up points, and operating times) and perceived accessibility

(individuals' perceptions of how well the service supported participation in daily and valued activities). Negative aspects of objective accessibility related mostly to the operating hours of the X-line. Several participants suggested extending operating hours to accommodate transportation on weeknights and weekends. For example, one participant expressed how the restricted operating hours impact on accessibility: *"*[…] *and run many more hours, so to speak, and even weekends, then my life would have been perfect"*, or as another participant described: *"Public transportation in Säffle has received poor ratings, especially when it comes to evening and weekend schedules. It doesn't affect me personally, but it does impact the youth. It affects access to the swimming pool, library, and other facilities."* Participants express that extended operating hours would benefit youngsters in particular, who may need access to these venues. Overall, participants were satisfied with the virtual pick-up and drop-off points, which were close in proximity to their homes. One participant mentioned*: "They drive up to the door if they have time, and help the elderly with everything"*, which adds to perceive accessibility. Another articulated great perceived accessibility and a possibility to go to where they want with the help of the service: *"*[…] *sometimes you can even get exactly where you want to go, so it's kind of like Uber."* Participants reflected that the service offered by the X-line supports their daily needs by adding simplicity, but also suggested improvements. For instance, one participant expressed that she has difficulties traveling when being on nightshifts since the X-line operating hours closed at 10 pm. Another user suggested that the X-line might also be used to transport larger goods (such as garbage bags and appliances) from point A to B, thus could provide accessibility to services beyond what traditional public transportation offers.

5.1.4. Perceived autonomy

Users' perceived autonomy was expressed in terms of freedom and the feeling of being able to choose when to travel. In this sense, autonomy seems closely linked to perceived accessibility. The more accessible an individual perceives a valued activity to be, the more likely they are to experience autonomy in pursuing the goal. The downsides of the X-line, such as limited operating times as already discussed, did not necessarily affect participants' perceived autonomy. Most participants mentioned a freedom to act within the service's limits, especially compared to regular public transport. However, the X-line could not compete with the freedom offered by a private car. Accordingly, those participants who were able to compare the freedom of car ownership with the X-line service reported a lower level of perceived autonomy than those who did not have this point of reference. As one participant said: *"I would have more freedom if I still had the car. But it's a sacrifice I chose to make for the sake of my wallet."* Regardless of car ownership, participants judged autonomy as an important facet of travel to various degrees. One participant

emphasized the importance of autonomy for immediate action and transportation in unexpected situations*: "I think it's pretty important. If you need to leave on short notice, it can be nice to take the X-line."* Another individual claimed it was important, but she had become accustomed to lacking autonomy, implying adaptability to current circumstances and transportation choices: *"Not full freedom. There are some days when you aren't able to travel late, and one time it happened that they stopped operating without any notice, so you have to adapt."*

Moreover, participants identified factors that affected their perception of autonomy, including aspects hindering autonomy. For example, one participant expressed that autonomy is extremely important, but needing to ask a neighbor for help significantly impacts the feeling of freedom, which is closely linked to autonomy. This suggests that independence may be a crucial factor for some people in their understanding of autonomy. Taken together, autonomy appeared more or less important in a practical sense, but also in a more abstract/philosophical sense, as an integral facet of living.

5.1.5. Technology

On-demand solutions are often connected to a phone application and the X-line is no exception to that rule, but it also provides users with a call-in option. The older generation experienced challenges and expressed strong wishes for improvements. For instance, one participant claimed she and others of her age do not know how to use the app or computer to make a booking, and therefore had difficulties accessing the service. Dissatisfaction with the call-in option were also mentioned, as staff seemed to lack knowledge of the X-line (the call-in was through a central service for the county). To enable ease of access, suggestions were made for a local booking service as well as staff who are educated on the X-line and its routes. Furthermore, another participant had no problems using the app but had complaints regarding the app itself, claiming the software was flawed and could not be used properly as intended. While a smart-phone app may be an essential feature of on-demand solutions, the opinions expressed by the elderly highlights the importance of finding solutions that caters for all age groups.

5.2. The provider's perspective on the DRT service

Thematic analysis of the interviews with DRT service providers yielded four main themes, eight 2nd order codes, and sixteen 1st order codes. The four main themes were identified as follows: (1) Provided autonomy, (2) Drivers enhancing the DRT experience, (3) Technology integration into DRT service provision, and (4) The dynamic nature of the service. Provided autonomy and Drivers

enhancing the DRT experiences are themes that stems from the interview guide, while Technology integration into DRT service provision and The dynamic nature of the service are all themes derived from spontaneous comments. The codes and aggregated dimensions/themes alongside participant quotes are presented in the table and discussed in more detail below.

The table above presents an overview of central quotations. The following sections offer a more detailed exploration of each main theme.

5.2.1. Provided autonomy

Across the interviewed service providers, a recurring theme was the increased freedom of choice and flexibility offered by the X-line adding to a sense of autonomy. All participants highlighted how the X-line provides users with the ability to travel almost whenever they desire, similar to the freedom of using a personal vehicle. This was strengthened in further discussions about the absence of fixed timetables and the ability to book rides on-demand. As resonated by one participant:

> *"Yes, so… the travel pattern has [changed], yes but before we had a city bus line that went its route and according to its timetable and it was fixed like that. It was the same route and the same times which means that the catchment area is very limited. Now there is no timetable, and there is no line route, which means that it is the customers themselves who decide when the bus should run and where it should go."*

The service providers explained how the X-line adjusts routes based on user requests which ensures that users have greater control over their travel plans. Consequently, service providers believe it enhances travelers' sense of autonomy in choosing when and where to travel, especially in comparison to traditional public transportation option. For example:

> *"We often jokingly say that public transport takes you from where you are not to where you do not want to go, but this service can at least take you very close from where you are, to very close to where you want to go. So it becomes a much more flexible and close solution than what public transport can be."*

Moreover, it was emphasized how the X-line empowers users to make independent travel decisions by providing a more personalized and door-to-door service. Service providers believe that the X-line allows users to tailor their travel experiences to their specific needs. One participant mentioned that the X-line users can travel directly to their desired destinations without the need for transfers or long waits, thus reducing reliance on factors such as timetables and route schedules:

> *"[…]the experience is that it is you who choose when the bus runs and to where it runs."* This increased autonomy in travel decisions contributes to a more seamless and efficient travel experience for users.

The service providers perceived that the X-line enhances users' autonomy by offering greater flexibility, freedom of choice, and independence in travel decisions. According to the participants, the DRT service empowers users to travel according to their own schedules and preferences, ultimately enhancing their overall travel experience and sense of autonomy.

5.2.2. Drivers enhancing the DRT experience

The service providers pointed to factors that may impact on the user experience of the X-line service, highlighting both positive and negative aspects. All participants mentioned how the commitment and enthusiasm of the drivers significantly enhanced users' overall experience of the X-line, often mentioning the experience in relation to security and familiarity:

> *"The drivers are very much appreciated, first of all, because of course it can be, it's a very important part that the drivers are positive and helpful and so, and we have that. It becomes a security that you recognize* [the driver]*, that you can say hello and that you get a hello back and you can exchange a few words. It becomes a great security for the passengers."*

The positive experience and sense of reassurance created by the driver also instills confidence in parents, allowing them to feel safe about letting their young children travel alone. As described by one participant:

> *"And they* [the drivers] *can see who has booked as well. It comes up in the booking, so they can say 'Hello Britta' or 'Hello Magnus' or, well, plus it's then the same passengers who travel more or less. Then you become, yes you become friends with them. So it becomes a security for parents who would let their children travel."*

These insights underscore the central role of drivers in shaping passengers' perceptions and overall satisfaction with the DRT service.

5.2.3. Technology integration into the DRT service provision

According to the service providers, the integration of technology within the X-line contributes to users' positive experience where the digital platform can foster a safe user experience. For example, as described by one participant: "*It's peak times where there might be waiting times, but the service itself and the system that we deliver and the foundation of the bookings, they are reliable, absolutely. And if there is something that doesn't work, we catch it right away. And that is a security*."

The participants especially highlighted the importance of the app's functionality, allowing users to effortlessly book and track their rides in real-time, and see this as a key factor in promoting service reliability and accessibility. For instance, it was emphasized how the dedicated app streamlines the booking process, thus enhancing overall user experience and ensuring greater

accessibility. As an example, users of the service do not need to step out of the comfort of their homes until the exact moment their ride is scheduled to arrive, thanks to the app providing real-time updates on service arrival times: "[…] *the bus arrives where I stand* […] *I don't stand by a bus pole, but I stand as close to home as possible*."

Additionally, the proximity to both pick-up points and final destinations significantly enhances accessibility for family members of children and the elderly. One participant gave an example of a son who felt secure letting his elderly mother travel alone with the X-line without needing to personally drive her to appointments:

> "*There was this elderly lady we picked up. Her son, who was probably my age, had booked the trip for her because she needed to run an errand in the city center. So, he didn't have to accompany her, meaning he didn't have to drive her or book a taxi, but rather he arranged a bus trip for her so he could continue with what he was doing instead of driving his mom or booking a more expensive taxi*."

However, there were some shortcomings with the app that could negatively impact accessibility for a particular group of users. As described by one participant: "*We've had some calls from customers who have, yes functional diversities, who felt uncertainty about when you don't know when the vehicle arrives if you don't' have the app*." Since real-time updates on travel require the use of the app, those without access may feel left out or unsure, potentially impacting their sense of inclusivity and negatively affecting their overall experience with the service.

5.2.4. The dynamic nature of the service

The service providers also highlighted how the X-line has shifted passengers' travel habits and redefined their approach to public transportation. One noticeable change is the expanded travel options and destinations served by the service. As one service provider pointed out: "*We now cover other areas in Säffle that weren't served by the previous city bus*." This expansion represents a departure from traditional fixed routes and schedules, enabling passengers to explore new parts of the town Säffle: "*All of a sudden, now you could travel all over the city, and in that way, partly, those trips, this travel-need exists and existed before as well. But now we created a service that made it possible*." This comment indicated that the X-line provided a solution to an unfulfilled travel demand and improved accessibility for users and to the city.

The X-line has led to a significant increase in ridership, attracting new passengers with diverse travel needs. As one service provider describes: "*we now have seven or even more times the number of passengers than before.*" This surge in ridership indicates the service's appeal to a broader user group within the local population, enhancing the overall accessibility of public transportation.

However, this success also brings a tradeoff; with more bookings come longer waiting times due to stretched capacity. Consequently, more passengers cancel their bookings and are possibly left feeling dissatisfied with the service decreasing both senses of autonomy and accessibility. One participant reflects on how this limitation has changed the expectations of public transport:

> *"Then I think it's interesting about Säffle that previously they had line 80 that ran once an hour, and then you get that service where you have a waiting time of 7, 8 minutes […], and then it became 20 minutes, and then people started canceling, it was too long of a wait. As a customer, you also become very spoiled very quickly."*

This comment indicates how individuals quickly become accustomed to higher service standards and quick accessibility and may feel dissatisfied when those standards are not met. To meet the strong demand and in response to longer waiting times, a second vehicle was introduced to address customer concerns and optimize service efficiency and accessibility.

The service's emphasis on accessibility and user autonomy empowers travelers to make independent choices about when and where they travel, drawing comparisons to the freedom associated with private car use. Despite some challenges, such as longer waiting times due to increased demand, the dynamic nature of the X-line and responsive adjustments by service providers demonstrate a commitment to meeting evolving passenger needs and enhancing overall service quality.

5.3. Comparing perspectives from service users and providers

In this section, we compare the perspectives of both the service providers and the users, focusing on the perceptions of autonomy and accessibility within the context of the X-line service. This comparison will provide insights into how the intentions of service provision align with the user experience (see Table 4).

Table 4. Summary of key findings for both user and provider perspectives.

Aspect	User perspectives	Service provider perspectives
Travel autonomy	Freedom to choose when and where to travel, but cannot fully match the convenience of a car.	Significant improvement compared to fixed routes, almost comparable to a car.
Interaction with drivers	Crucial for overall experience and sense of security.	Recognized as important for customer satisfaction.
Perceived accessibility	Appreciated and acknowledged, but affected by certain limitations.	Expanded coverage increases the number of travelers.
Technology	Concerns about the app's usability for older users, although no significant issues reported.	No significant issues reported.
Service challenges	Longer wait times affect the experience.	Demand exceeds capacity.

The following sections discuss the dual perspectives on travel autonomy and perceived accessibility.

5.3.1. Dual perspectives on travel autonomy

From the users' standpoint, autonomy was perceived as closely intertwined with the freedom to choose when to travel. Despite some limitations regarding operating times, participants felt a sense of freedom provided by the service. While some acknowledged that the X-line could not replicate the freedom offered by car ownership, they still highlighted its role in providing autonomy, particularly when not having access to a private car. Service providers, on the other hand, emphasized how the X-line offers a service that closely mirrors the freedom of using a personal vehicle. This suggests a discrepancy between the providers' intentions of the service's impact on users' perceived autonomy. Users emphasized the importance of autonomy and expressed that having to depend on others for assistance can reduce feelings of independence. This indicates that independence is a crucial factor in how some individuals perceive their autonomy. In alignment with this perspective, service providers suggested that the increased ridership among certain age groups, like the elderly, may suggest that the accessibility and convenience provided by the X-line reduce the need to depend on others for transportation, thereby enhancing autonomy and independence. Overall, the alignment between users' and providers' perspectives suggests that the X-line effectively supports experiences of autonomy in travel. While users may adapt to the service's limitations, such as operating times, the flexibility and freedom offered by the X-line contribute significantly to enhancing users' autonomy and overall travel experience.

Service providers emphasized the increased freedom of choice and flexibility through the X-line compared to conventional public transportation. This included the absence of fixed timetables and the ability to customize routes, which empower users to make independent travel decisions, fostering travel autonomy. This perspective aligns with users' experiences, as they claim that the service supports their daily needs by adding simplicity. Users expressed their experience of autonomy in terms of being able to choose when to travel, which the X-line supports.

5.3.2. Dual perspectives on perceived accessibility

While service providers underscored the increased freedom offered by the X-line compared to traditional public transportation, users expressed concerns about increasing waiting times and decreasing sense of access. This indicates a gap between users' subjective measures of accessibility and flexibility and service providers' perceived support for individual needs.

Another factor affecting the perception of accessibility is the users' technology readiness and the ability to use the online platform to access and book the travel service. Here, the app's functionality played a crucial role in shaping the user experience. Users appreciated the ability to specify both preferred arrival and/or pickup times and places when booking, which contributed to both a sense of control and accessibility. Service providers placed more focus on the real-time updates through the app as enhancing sense of security. While service providers emphasized the benefits of the digital platform and real-time updates, users got frustrated by the real-time updates on arrival time which many times are moved back due to delays.

There was also a division in the experience of using the online services (especially the app as the central booking and information platform) among different user groups. Older users struggled with using the app and related online services. Here, the service providers made no such distinction but were generally optimistic about the app's user-friendliness and accessibility for all age groups. This optimism ignores the specific needs of older or less tech-savvy users, negatively impacting their perceived accessibility. Ensuring that the app is equally accessible and user-friendly for all age groups would help bridge this gap and enhance the overall experience of the service, particularly in relation to accessibility and autonomy.

6. Discussion

This study used the piloting of a Demand Responsive Transport (DRT) service, replacing a traditional public transportation service, as a case to explore adult service users' perceptions of travel autonomy and accessibility. In addition, the study investigated how the service provider's intention of providing autonomy and accessibility aligns with the users' perceptions. By explicitly combining user and service provider perspectives, the study offers a dual-perspective analysis of autonomy and accessibility in a rural DRT context, addressing a gap in existing research that has largely examined these issues from either a user or a system perspective, but rarely both simultaneously. As autonomy and accessibility are intertwined concepts (Støre et al., 2025), we discuss these concepts in combined form to draw relevant implications for DRT services and their contribution to sustainable mobility and community livability. We further explore the (dis)alignment between service providers' perceptions of the service and users' experiences, emphasizing how closing this gap can enhance both service effectiveness and user well-being.

From the users' perspective, autonomy is linked to the freedom and flexibility to choose when to travel. Although the DRT service had limited operating hours, we found that users still experienced a sense of freedom and feeling of autonomy. It seems that the ability to choose when to travel is a crucial aspect of a user's feeling of autonomy. This perception is especially the case for individuals

without access to a private car, who might otherwise feel restricted by traditional public transportation schedules. The DRT service studied allowed for route customization and flexible scheduling, which was seen as a significant improvement over fixed-route and timetable-bound public transit options, and different from some DRT solutions using pre-booking by telephone (e.g. Gkavra et al., 2024). Even though a DRT service does not fully replicate the on-demand nature of car ownership, it still provides a notable degree of independence. This is particularly so for individuals with lower income, the elderly, and individuals with impairments (also confirmed in Bascom & Christensen, 2017 and Márquez et al., 2019). By offering a more adaptable, accessible, and user-friendly service, a DRT service such as the X-line helps bridge the mobility gap, ensuring a more diverse user group has access to essential services and social opportunities, thereby contributing to community livability and quality of life.

From the provider's perspective, DRT services were intended to offer freedom comparable to private car use. Providers emphasized flexibility and convenience, believing that users could travel almost as freely as they would with a private vehicle. This indicates that providers primarily conceptualize autonomy at a system-design level, whereas users experience autonomy at the level of everyday control and reliability. This divergence between service providers' intentions and users' lived experiences highlights the analytical value of the dual perspective adopted in this study, revealing how autonomy and accessibility are operationalized differently in planning *versus* everyday use. While providers viewed the X-line as a comprehensive and fully flexible transportation solution, users were more aware of its limitations, such as restricted operating hours and occasional waiting times. This gap highlights the importance of aligning service design with realistic expectations of autonomy and accessibility. In rural contexts, such misalignment may lead to overestimating DRT's ability to substitute private car use, a relevant observation contributing to the mixed evidence on car dependence and ownership (Schasché et al., 2022).

A key implication is that enhancing perceived independence should be central to DRT design. This includes ensuring digital accessibility through online or mobile booking systems that accommodate vulnerable groups such as older adults. In our study, the mobile app's functionality strongly shaped perceived accessibility, echoing earlier findings on the role of booking systems in mode choice (Schasché et al., 2022). Users emphasized the importance of specifying preferred arrival and pickup times, which enhanced their sense of control and accessibility. Service providers, in contrast, focused more on the real-time updates through the app, which they believed enhanced safety and perceived reliability. However, users sometimes experienced these updates as frustrating when delays were communicated, revealing a mismatch in perceived value. Moreover, a generational divide emerged: younger adult users reported ease of use, while older users struggled with the app's functionality. Designing DRT services to be accessible and user-friendly is therefore essential to enhance

perceived accessibility, social inclusion, and the overall livability benefits of such systems.

Service providers highlighted that DRT offers greater freedom and flexibility than conventional public transportation. The absence of fixed timetables and the ability to customize routes empower users to make independent travel decisions, fostering a sense of autonomy through increased accessibility. This viewpoint aligned with the user experience, as users appreciated the ability to choose their travel times. However, there were still discrepancies regarding perceived accessibility. Users valued flexibility but reported concerns about waiting times and uncertainty, particularly during peak periods. This finding resonates with studies showing that users value the flexibility but remain sensitive to reliability and availability (Liyanage et al., 2019). Unexpected waiting times can reduce user satisfaction negatively impact perceptions of service quality and well-being.

A unique contribution of our study is that DRT services, such as the X-line, appear to be suitable substitutes for regular public transportation in rural or small-town contexts, where fixed timetables make traditional public transportation ineffective and unsustainable. By combining user and service provider perspectives, the study further demonstrates how perceived autonomy and accessibility are shaped not only by service design intentions but also by everyday use and interaction with the system. Well-designed DRT systems enhance accessibility and social participation for many users, improving the livability of less densely populated regions. However, DRT may not serve as an effective replacement for those who usually travel by car, particularly in rural areas, where private cars remain the most convenient option. This indicates that while DRT can enhance mobility equity and social inclusion, its potential to reduce car dependence in rural areas may be limited.

Although older adults are often highlighted as a key target group for demand-responsive transport services, the present study does not treat older users as a distinct analytical subgroup. Instead, age is approached as one contextual factor among others shaping user experiences, and insights related to older users should therefore be understood as indicative rather than representative. This reflects the limited number of participants over 60 in our sample. Furthermore, younger age groups were not included in the study. We acknowledge this as a limitation, as these groups may have other needs and perceptions of DRT services than those identified in the interviews. This has implications for the transferability of the findings, which should be understood as analytically transferable to similar rural demand-responsive contexts rather than statistically representative of the full user population. Additional research is therefore warranted, with particularly attention to school children, adolescents, and younger adults.

The driver emerged as a key factor shaping both perceived security and social connectedness, two crucial components of transport-related livability. Positive interactions with drivers enhanced users' comfort and willingness to reuse the

service, while negative experiences discouraged future use. The driver thus contributes not only to perceived safety but also to emotional and social well-being. This human element, which reinforces feelings of perceived autonomy and accessibility, is highly relevant for discussions about the future of automated mobility. As autonomous vehicles and driverless systems evolve, it will be important to consider how to replicate the sense of social reassurance currently provided by human drivers.

Taken together, the findings allow us to identify a set of key enablers and barriers shaping perceived autonomy and perceived accessibility in rural DRT use. Enablers include flexible on-demand booking, close pick-up and drop-off locations, supportive driver–passenger interaction, and real-time travel information when accessible to the user. Barriers include limited operating hours, capacity constraints, and longer waiting times, dependence on digital booking tools, and mismatches between provider assumptions and everyday user experience. These factors indicate that DRT design features and operational conditions directly influence whether users experience the service as autonomy-supporting and accessible. Future research should examine how these enabling and constraining factors vary across user groups, levels of digital readiness, and rural service configurations, and how they relate to long-term adoption and substitution of private car travel.

7. Conclusion

In conclusion, the present case study on the DRT service "the X-line" piloted in a rural town in Sweden, provides new insights into perceptions of autonomy and accessibility both by those using and those providing the service. The study highlights how such services can support sustainable and inclusive mobility by offering greater flexibility and independence than traditional public transport, thereby contributing to community livability and perceived well-being. In rural contexts, these experiential factors are particularly important because they shape whether DRT can function as a credible alternative to private car travel for at least some trip purposes.

While DRT services enhance a sense of travel autonomy and accessibility, they do not provide the level of flexibility and autonomy as private car travel. The findings also underscore the importance of user-centered service design, particularly in developing an easy-to-use online platform and ensuring inclusive digital access. Moreover, the study reaffirms the critical role of drivers as both service ambassadors and enablers of perceived safety and accessibility.

By identifying areas of alignment and mismatch between user experiences and provider intentions, this study contributes to a more nuanced understanding of how autonomy and accessibility are realized in practice. These insights have direct policy relevance for planners and decision-makers seeking to design sustainable transport interventions that enhance livability and equity across

different geographical contexts. Although based on a single case, the results offer valuable implications for the development of flexible mobility services in other rural and urban areas, and future comparative research is encouraged to explore context-specific variations related to technology readiness, cultural norms, and transport system maturity (Polydoropoulou et al., 2020). Limitations related to the small number of older participants should be considered when interpreting findings related to this group, and the absence of children, youth, and young adults warrants further research.

Disclosure statement

No potential conflict of interest was reported by the author(s).

Disclosure of Generative AI assistance

Generative AI (ChatGPT, GPT-5 version by OpenAI, 2025) was used to assist with language editing and to verify reference formatting. It was also used for coding assistance, specifically to support the organization and refinement of initial codes during the thematic analysis process. All coding decisions, interpretations, and thematic categorizations were made and verified by the authors. The authors ensure that the scientific content and interpretations are entirely their own.

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